



Daffodil
International
University

ASSIGNMENT

Course Code: CIS211

Course Title: Computer Networks

Topic Name: Public Wi-Fi and Guest Network Design for a Shopping Mall

Submitted To

Name: Safayat Hossain

Designation: Lecturer

Department of Computing and Information System

Daffodil International University

Submitted By

Name: Md Ratul Hasan Abdullah Bin Wahed
..... Samiul Islam Tamim MD. Amzedullah
Ratul

ID: 252-16-012 252-16-016 252-16-035
..... 252-16-048

Section: 22A

Semester: 4th

Department of Computing and Information System

Daffodil International University

Date of Submission: 10/08/2026

Table of Contents

- **Introduction**
-
- **Project Objectives**
- **Scenario Description**
- **Network Architecture and Topology**
- **OSI Model, Network Type, Architecture and Topology**
- **Network Protocols**
- **IP Addressing, Subnetting and VLAN**
- **Routing Protocols**
- **Application and Network Protocol Analysis**
- **Hardware, Media and Connectors**
- **Wireless Network Evaluation**
- **Network Security Evaluation**
- **Overall Evaluation**
- **Cisco Packet Tracer Implementation**
- **Testing and Verification**
- **Troubleshooting**
- **Future Improvements**
- **Limitations**
- **Conclusion**
- **References**
- **Annex A: AI Use Disclosure**

1. Introduction

Shopping malls are increasingly dependent on computer networks to support both their internal operations and customer-facing digital services. One of the most common services expected by customers is public Wi-Fi access. A well-designed public Wi-Fi network allows visitors to connect their smartphones, laptops, and other wireless devices to the Internet while employees continue to use separate network resources for their daily activities.

This project presents a Public Wi-Fi and Guest Network Design for a Shopping Mall developed and simulated using Cisco Packet Tracer. The network is designed to provide Internet connectivity to mall visitors while also supporting different operational areas such as management, shops, food court, and security.

The proposed network consists of a Mall_Router, a central Core_Switch, several dedicated departmental switches, a Main_Server, a dedicated Guest_SW, and a wireless access point named MallGuest_AP. Wired computers are connected to the appropriate departmental switches, while guest users connect wirelessly through the public Wi-Fi access point.

The overall design focuses on organized network connectivity, centralized switching, wireless access, network segmentation, security, reliability, and future scalability.

2. Project Objectives

The main things we want to do in this project are:

To make a Wi-Fi network for a shopping mall.

To give people who visit the mall access to the internet.

To create a network that works well for all the parts of the mall.

To connect the management team, shops, food court and security devices to the network.

To use a Core Switch to connect all the parts of the network

To connect the mall network to the internet using the Mall_Router.
To make a Wi-Fi network just for guests.
To organize the network devices so they make sense for what they do.
To set up the addresses for the devices on the network.
To look at how to divide the network into parts using VLANs.
To figure out which routing protocols are best.
To find the hardware and equipment for the network.
To see how well the wireless network covers the mall and how many people can use it.
To find out what security risks there are with Wi-Fi.
To test the network using Cisco Packet Tracer.
To design a network that can grow and change in the future.

3. Scenario Description

The network we are making is for a shopping mall that gives Wi-Fi to customers and has a separate network for the people who work there.

The mall has a lot of areas including:

- Management Area
- Shopping Area
- Food Court
- Security Area
- Public Guest Wi-Fi Area
- Central Server/Network Area

The management area has computers like Admin_PC and Manager_PC. The shop area has a computer called Shop_PC and the food court has a computer called FoodCourt_PC. The security area has a computer called Security_PC.

We are making a Wi-Fi network just for guests. The MallGuest_AP is connected to Guest_SW, which lets devices, like Guest_Laptop_1, Guest_Laptop_2, Guest_Phone_1 and Guest_Phone_2 connect to the internet.

All the different areas of the mall are connected to a Core_Switch, which is the main part of the network.

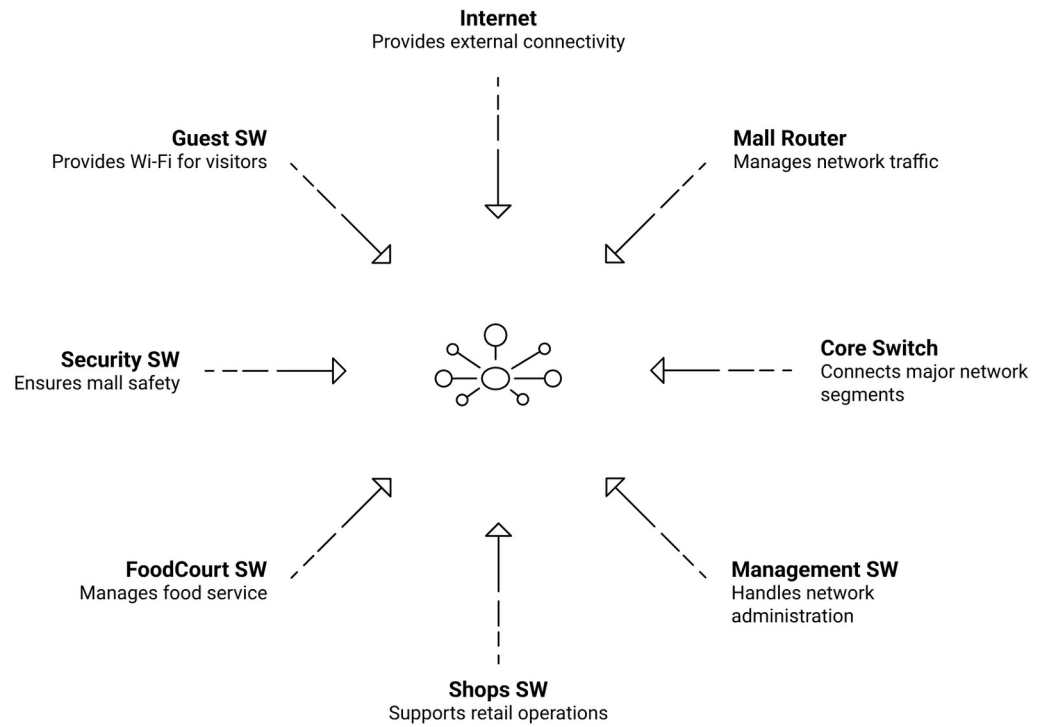
The Main_Server is also connected to the Core_Switch. The Mall_Router connects the mall network to the internet.

4. Network Architecture and Topology

The network follows a centralized architecture in which the **Core_Switch** acts as the main switching point.

The major structure of the network is:

Mall Network Infrastructure



Made with  Napkin

.The **Main_Server** is directly connected to the **Core_Switch**.

Network Components and Functions

Component	Device Name	Function
 Router	Mall_Router	Internet/network routing
 Core Switch	Core_Switch	network connectivity
 Switch	Management_SW	Management devices
 Switch	Shops_SW	Shop network
 Switch	FoodCourt_SW	Food court network
 Switch	Security_SW	Security network
 Switch	Guest_SW	Public Wi-Fi network
 Server	Main_Server	Central server/network services
 Access Point	MallGuest_AP	Guest wireless connectivity

This design can be described as an **extended star topology**, because multiple switches and network segments are connected through a central switching device.

5. OSI Model, Network Type, Architecture and Topology

5.1 OSI Model

The Open Systems Interconnection (OSI) model provides a framework for understanding how data moves through the shopping mall network.

OSI Layer Application, in This Project

Layer 7 – Application Web, DNS, DHCP and other network services

Layer 6 – Presentation Data formatting and encryption

Layer 5 – Session Establishing and maintaining communication sessions

Layer 4 – Transport TCP and UDP communication

Layer 3 – Network IP addressing and routing

Layer 2 – Data Link MAC addressing and Ethernet/Wi-Fi frames

Layer 1 – Physical Ethernet cables and wireless signals

For example when a guest uses Guest_Laptop_1 to access the Internet the data travels from the laptop through the MallGuest_AP, Guest_SW, Core_Switch and Mall_Router before reaching the Internet.

Here is the rewritten version of the input text:

5.2 Network Type

The main type of network is a Local Area Network (LAN) because the network setup is placed inside the shopping malls area.

The part for guests is a Wireless LAN (WLAN) because people use devices to connect.

5.3 Network Architecture

The project uses a network setup.

The Core_Switch is the point where everything connects and other switches link to different parts of the building.

This helps keep the network easier to run and take care of.

5.4 Network Topology

The way the network is set up is called an Extended Star Topology. The Core_Switch is in the middle and many other switches connect to it.

The main benefits are:

- Easy to manage
- Easy to fix problems
- Simple to add parts
- Neat and clear network layout
- If something goes wrong it affects only one part of the network
- Network Protocols
- Several network protocols are important for the shopping mall network.

Internet Protocol (IP)

IP gives each device an address.

Every device needs the IP address so that data can get to the right place.

Transmission Control Protocol (TCP)

TCP makes sure that data is sent correctly and in order.

It is good for apps where it's important that the information arrives without any problems.

Dynamic Host Configuration Protocol (DHCP)

DHCP gives network settings to devices automatically.

This is great for Wi-Fi because people connect and leave often.

Of setting up each guest device by hand DHCP can give the needed information on its own.

Domain Name System (DNS)

DNS changes website names into numbers that devices can understand.

This means people can use website names of remembering long numbers.

HTTP/HTTPS

HTTP and HTTPS are types of protocols used for web traffic.

HTTPS is important for Wi-Fi because it keeps the connection, between users and websites safe and secure.

Network Configuration Details

Network Area	Devices	VLAN	Network Address	Subnet Mask	Usable Host Range	Broadcast Address	Default Gateway
 Management	Admin_PC, Manager_PC	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]
 Shops	Shop_PC	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]
 Food Court	FoodCourt_PC	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]
 Security	Security_PC	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]
 Guest Wi-Fi	Guest_Laptop_1, Guest_Laptop_2, Guest_Phone_1, Guest_Phone_2	[Actual]	[DHCP Network]	[DHCP Mask]	[DHCP Assigned Range]	[Broadcast]	[DHCP Gateway]
 Server	Main_Server	—	[Actual]	[Actual]	[Actual]	[Actual]	[Actual]

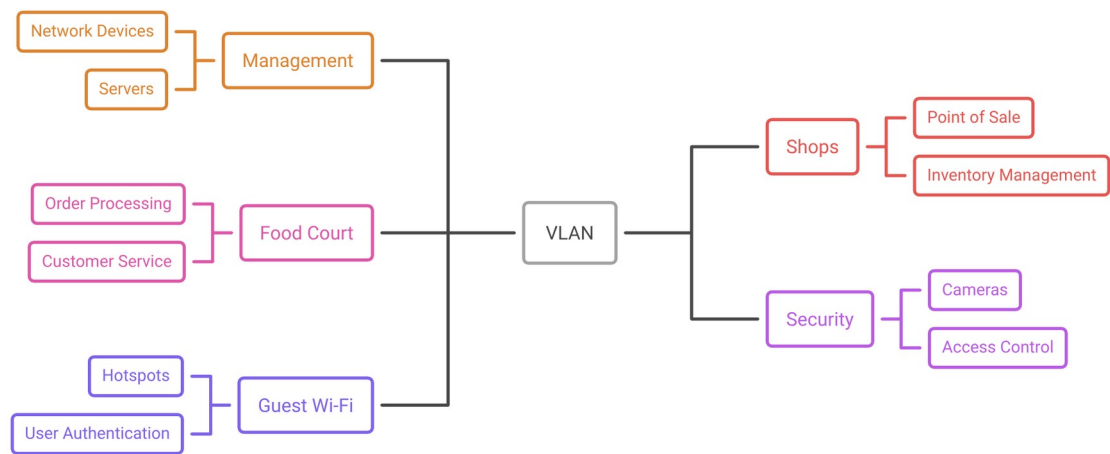
Made with  Napkin

.

VLAN Segmentation

VLAN segmentation can be used to logically separate the different

VLAN Structure in a Network



Made with Napkin

network areas.Like:

.

8. Routing Protocols

Routing is necessary when communication is required between different IP networks.

Two commonly considered routing protocols are **RIP** and **OSPF**.

Comparison of RIP and OSPF Routing Protocols

Feature	Type	Metric	Convergence	Scalability	Configuration	Suitable for
RIP	Distance Vector	Hop Count	Relatively slower	Limited	Easier	Smaller networks
OSPF	Link State	Cost	Faster	Better	More advanced	Medium/Large networks

Made with Napkin

.

9. Application and Network Protocol Analysis

9.1 DHCP

The DHCP protocol is really useful for the Guest Wi-Fi network.

When customers come to the mall they are there for a short time so it would be a waste of time to assign an IP address to every device manually.

The DHCP protocol can give devices the following information:

* IP address * Subnet mask * Default gateway * DNS information

9.2 DNS

The DNS protocol makes it easy for people to access the Internet by turning domain names into IP addresses.

This means a guest can just type in a website name of having to know the IP address of the server they want to visit.

9.3 HTTPS

The HTTPS protocol keeps web communication safe by encrypting data between the client and the web server.

For people using Wi-Fi it is very important to have encrypted communication because there are many unknown users on the wireless network.

Protocol Comparison

- Protocol Main Purpose Reliability Security Importance
- DHCP IP configuration High Network access control
- DNS Name resolution High DNS security is important
- HTTPS Secure web communication High Encryption

10. Hardware, Media and Connectors

The project has different types of networking hardware.

10.1 Mall_Router : The Mall_Router is the router for the network. It connects the mall network to the Internet.

10.2 Core_Switch: The Core_Switch is the switching device. It is connected to the following devices:

* Management_SW * Shops_SW * FoodCourt_SW * Security_SW * Guest_SW * Main_Server * Mall_Router

So the Core_Switch is the point of the network topology.

10.3 Management_SW

This switch gives connectivity to the following devices:

* Admin_PC * Manager_PC

10.4 Shops_SW: This switch gives connectivity to Shop_PC.

10.5 FoodCourt_SW: This switch gives connectivity to FoodCourt_PC.

10.6 Security_SW: This switch gives connectivity to Security_PC.

10.7 Guest_SW: The Guest_SW is used for the public Wi-Fi section.

It connects the MallGuest_AP to the rest of the network.

10.8 Main_Server: The Main_Server is connected directly to the Core_Switch. Is the centralized server infrastructure of the mall.

10.9 MallGuest_AP: The wireless access point gives Wi-Fi connectivity to mall visitors.

The project includes the following wireless client devices:

* Guest_Laptop_1 * Guest_Laptop_2 * Guest_Phone_1 * Guest_Phone_2

10.10 Network Media: The wired infrastructure uses Ethernet connections between routers, switches, servers and PCs. Wireless communication is used between the MallGuest_AP and guest devices.

11. Wireless Network Evaluation

The main feature of this project is wireless connectivity. The MallGuest_AP gives connectivity to visitors in the mall. The wireless network allows many types of devices to connect, including:

* Smartphones * Laptops * Tablets and similar wireless clients

Access Point Placement

The Access Point should be placed in a public area where it can give suitable coverage to the largest possible number of users. In a shopping mall you would need more Access Points for large areas or locations, with many people.

Capacity: The wireless network should consider the following things:

* Number of users * Available bandwidth * AP capacity * Signal strength * Interference * Physical obstacles

Wireless Security: The guest wireless network should use an authentication and encryption mechanism to keep the network safe.

12.Network Security Evaluation

When people use Wi-Fi it is not very safe because we do not know who the users are and we cannot trust them.

12.1 Unauthorized Access

Someone might try to get into the malls network through the guest network. To stop this from happening we should keep guest users separate from the networks using something called VLANs and access-control policies.

12.2 Man-in-the-Middle Attack

Someone might try to intercept what people are sending and receiving on the network. To prevent this we should use protocols like HTTPS and make sure the wireless network is safe.

12.3 Data Interception

If people are sending information without encrypting it someone might be able to see what they are sending. To stop this from happening people should use encrypted services and the wireless network should have security mechanisms.

12.4 Denial of Service

If there is much traffic on the network it can stop people from being able to use it. To prevent this we should keep an eye on the traffic manage the bandwidth and have security controls.

12.5 Guest Network Isolation

The Guest network should be kept separate from parts of the network like:

- Management

- Security
- Server
- Other sensitive internal resources

This is one of the important things to keep in mind when designing the network.

13.Overall Evaluation

The network we have designed is good for a shopping mall because it allows people to connect to the Internet. It also keeps the internal network safe. The Core_Switch is the part of the network and there are separate switches for different parts of the mall. The Guest_SW and MallGuest_AP provide a path for people using the public Wi-Fi. To make the network even safer we can use VLAN segmentation and access-control policies to keep guest users from getting into internal resources. We can also add switches, access points and network services to make the network bigger. The network is easy to manage and fix if something goes wrong. In the future we could add more redundancy to make it even better. To keep the network safe we can use things like network segmentation, wireless security, ACLs and firewall policies.

14.Cisco Packet Tracer Implementation

We used Cisco Packet Tracer to design and test the network.

The network has:

1. One Internet connection
2. One Mall_Router
3. One Core_Switch
4. Five access switches
5. One Main_Server
6. One wireless access point
7. Wired client PCs
8. Wireless guest devices

14.1 Complete Topology

Figure 1: Complete Public Wi-Fi Mall Network in Cisco Packet Tracer

This is the part of the network that people use when they visit the mall.

15. Testing and Verification

We need to test the network to make sure everything is working correctly.

15.1 Ping Testing

We can use the ping command to see if devices can talk to each other. For example we can use the command: ping [Destination IP]. If we get a response it means the devices can talk to each other.

Figure 2

connectivity test between network devices

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Failed	Shop_PC	FoodCourt_PC	ICMP		0.000	N	0
	Successful	Shop_PC	FoodCourt_PC	ICMP		0.000	N	1

15.2 Interface Verification

We can use the command "show ip interface brief" on the router to see:

- Interface IP addresses
- Interface status
- Protocol status

Figure 3

Router Interface Verification

15.3 Routing Verification

We can use the command "show ip route" to see the routing table of the router.

Figure 4

Mall_Router Routing Table

15.4 VLAN Verification

If we have set up VLANs we can use the command "show vlan brief" to see:

- VLAN IDs
- Switch port assignments

Figure 5

VLAN Configuration Verification

15.5 Simulation Mode

We can use Cisco Packet Tracer Simulation Mode to see how packets move through the network.

The simulation can show us how a packet travels from a guest device to the Internet.

Figure 6

Packet Flow Demonstration in Simulation Mode

16.Troubleshooting

Sometimes the network might not work correctly because of problems with the configuration or the devices. We need to troubleshoot the network to find and fix the problems.

16.1 Connectivity Troubleshooting

We can follow these steps to troubleshoot connectivity problems:

1. Check the connections.
2. Check the interface status.
3. Verify the IP addresses.
4. Verify the subnet masks.
5. Check the default gateways.
6. Verify the routing configuration.
7. Check the VLAN configuration.
8. Test connectivity using ping.

16.2 Routing Troubleshooting

We can use the command "show ip route" to check the routing table. If a network is missing we need to review the routing configuration.

16.3 VLAN Troubleshooting

We can use the command "show vlan brief" to verify the VLAN configuration. If the VLAN assignments are incorrect devices might not be able to talk to each other.

16.4 Wireless Troubleshooting

For guest connectivity we should check:

- Access Point status
- Wireless configuration
- Guest device connection
- IP configuration
- Default gateway
- Network reachability

17.Future Improvements

We can improve the network in ways.

17.1 Additional Access Points: Big shopping malls need multiple access points to provide coverage.

17.2 Portal: We can set up a captive portal to make guests accept terms and conditions or authenticate before using the Internet.

17.3 Firewall: We can add a firewall to protect the guest network from the internal networks.

17.4 ACL: We can use Access Control Lists to restrict guest traffic from reaching internal resources.

17.5 Network Monitoring: We can set up a monitoring system to detect:

- High traffic
- Device failures
- Unauthorized activity
- Network performance issues

17.6 Redundancy: We can add more routers and switches to reduce single points of failure.

17.7 Better Wireless Capacity: We can add access points and use modern Wi-Fi technologies to increase the wireless capacity.

18. Limitations

The network we designed has some limitations. First it is a simulation, not a network.

In the world wireless coverage depends on:

- Building structure
- Walls
- Interference
- AP placement
- User density
- Hardware capabilities

The network we designed is also smaller than a shopping mall network.

A real network would need to consider things like ISP bandwidth, firewall hardware, wireless controllers, authentication systems, monitoring, redundancy, cabling and maintenance costs.

19. Conclusion

The Public Wi-Fi and Guest Network Design for a Shopping Mall provides a network solution for supporting both public and internal connectivity. The network uses a Core_Switch connected to dedicated switches for management, shops, food court, security and guest connectivity.

The Mall_Router provides the connection to the Internet. The Main_Server represents the centralized server infrastructure. The public Wi-Fi section is implemented using Guest_SW and MallGuest_AP which provide connectivity to guest laptops and smartphones.

The proposed architecture provides a foundation for network organization and can be improved further through VLAN segmentation, ACLs, firewall policies, stronger wireless security, additional access points, monitoring and redundant infrastructure.

Overall the project shows how a shopping mall can combine wired and wireless networking technologies to provide Internet access while maintaining a structured and secure internal network.

20. References

Daffodil International University, Department of Computing and Information System CIS211 Computer Networks Semester Assignment.

Cisco Systems, Cisco Packet Tracer.

Cisco Networking Academy, Introduction, to Networks.

Internet Engineering Task Force (IETF) Internet Protocol standards.

IEEE, IEEE 802.11 Wireless LAN Standards.

21. Annex A: AI Use Disclosure

AI Tool Used

ChatGPT – GPT-5.6 Luna, Napkin Ai

Purpose of AI Use

AI assistance was used for:

- Organizing the report structure.
- Improving grammar and presentation.
- Explaining general networking concepts.
- Helping organize the technical documentation of the project.
- To make the workflow visuals